Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/29/2007
PI Name:	Young, Gerald W Ph.D.		
Project Title:	Theory, Modeling, Software and Hardware Development in Computational Materials Science		
Division Name:	Physical Sciences		
Program/Discipline:	OTHER		
Program/Discipline--Element/Subdiscipline:	OTHER--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Akron	**State:**	OH
Zip Code:	44325-4002	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/07/2004	**End Date:**	03/03/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chait, Arnon	**Contact Phone:**	216-433-3558
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Clemons, Curtis (University of Akron) HARIHARAN, S.I. (University of Akron)		
Grant/Contract No.:	NNC04GB27G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The focus of this Cooperative Agreement between the Computational Materials Laboratory (CML) of the Processing Science and Technology Branch of the NASA Glenn Research Center (GRC) and the Department of Theoretical and Applied Mathematics at The University of Akron is in the areas of system development of the CML workstation environment, modeling of microgravity and earth-based material processing systems, and joint activities in laboratory projects. These efforts complement each other as the majority of the modeling work involves numerical computations to support laboratory investigations.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The modeling and simulation effort of nanofiber deposition and the subsequent behavior of coated nanofibers and nanotubes made progress toward answering the following fundamental questions regarding nanotube synthesis and processing, as well as thin film coating in general: (1) How can one obtain a uniform coating thickness around the outside diameter of the sacrificial fiber? (2) In plasma deposition processes, a sheath region forms near the substrate. The ions accelerate through the sheath and bombard the substrate. How do the fiber diameter and the sheath thickness affect the coating uniformity? (3) How does the roughness of the coating depend upon the reactor process variables? (4) How does the coating thickness and roughness affect the electromagnetic properties of the tube? In the area of lithographic patterning techniques based on scanned probe oxidation we provided fundamental mathematical tools and scientific understanding needed for this critical deposition technology. We developed, solved, and analyzed models for scanned probe oxidation as a means for writing oxide patterns on a substrate. The model includes phase boundaries, reaction-diffusion processes, electric fields, and thermal and mechanical oscillations within the system. Free boundaries separate layers of substrate, oxide and fluid regions. The coupling between the layers influences the mechanical and reactive mechanisms, ion and electron transport and electromagnetic fields in the system. The models developed for this process broaden the mathematical formalism needed to address similar types of systems. Further these models are being used to increase the uniformity of nano patterning and will ultimately be applied in process control strategies.
Task Progress:	The original grant proposal was for the three year period, 6/7/04 - 3/3/07. Note that due to NASA budget constraints only the first year of the proposal was funded. Hence, $128,000 of the requested $395,636 for the cooperative agreement was funded. Due to the decrease in budget it was only possible after 10/31/05 to support University of Akron personnel to work on site at NASA GRC part-time during the period June 12, 2006 to August 6, 2006. Nevertheless the investigative team continued work at The University of Akron. During the time period at NASA GRC, the system management team investigated, recommended, and maintained state-of-the-art equipment such as workstations, printers, disk drives, memory, graphics cards, monitors, CPU upgrades, keyboards, ethernet cards, and tape back-up systems. The team managed network services including printing, user accounting, server management, hardware and software installation, backups, and data recovery. Software packages were maintained. Additionally, the team investigated and evaluated new software and hardware as these became available, making recommendations as to which should be acquired to enhance the capabilities of the CML at NASA GRC. Modeling projects at The University of Akron concerned the development and application of mathematical methods for the formulation, analysis and solution of material processing system models. Specifically, we examined nanoscale models of scanned probe oxidation, and models for the coating and applications of nanofibers. In general the team examined simple model problems to determine the process sensitivity to the operating conditions, describe the scientific phenomena of heat, mass, and momentum transport within the system, and elucidated the coupled dynamics of these transport mechanisms. A prior cooperative agreement, NASA Grant NCC3-1094, established foundation work in these areas. The team continued working in these areas. Hence, the team considered: (1) development of an investigation that coupled mathematical modeling and analysis to study lithographic patterning techniques based on scanned probe oxidation at the nanoscale. (2) development of models for the investigation of the coating of polymer nanofibers with conducting materials, and the subsequent response of the coated nanofibers (or newly formed nanotubes after dissolution of the nanofiber core) to applied electromagnetic fields.
Bibliography Type:	Description: (Last Updated: 04/03/2007)
Articles in Peer-reviewed Journals	Orians A, Clemons CB, Golovaty D, Young GW. "One-dimensional dynamics of nano-scale oxidation." Surface Science, 2006 Aug 15;600(16):3297-312. http://dx.doi.org/10.1016/j.susc.2006.06.017 , Aug-2006
Articles in Peer-reviewed Journals	Moore K, Clemons CB, Kreider KL,Young GW. "Modeling and simulation of axisymmetric coating growth on nanofibers." Journal of Applied Physics, 2007 Mar 15;101(6):064305-12 pgs. http://dx.doi.org/10.1063/1.2710442 , Mar-2007
Articles in Peer-reviewed Journals	Buldum A, Busuladzic I, Clemons CB, Dill LH, Kreider KL, Young GW, Evans EA, Zhang G, Hariharan SI, Keifer W. "Multiscale modeling, simulations, and experiments of coating growth on nanofibers. Part I. Sputtering." Journal of Applied Physics, 2005 Aug 15;98(45):044303-10 pgs. http://dx.doi.org/10.1063/1.2007848 , Aug-2005
Articles in Peer-reviewed Journals	Buldum A, Clemons CB, Dill LH, Kreider KL, Young GW, Zheng X, Evans EA, Zhang G, Hariharan SI. "Multiscale modeling, simulations, and experiments of coating growth on nanofibers. Part II. Deposition." Journal of Applied Physics, 2005 Aug 15;98:044304-16 pgs. http://dx.doi.org/10.1063/1.2007849 , Aug-2005
Articles in Peer-reviewed Journals	Marinov T, Buldum A, Clemons CB, Kreider KL, Young GW, Hariharan SI. "Field emission from coated nanowires." Journal of Applied Physics, 2005 Aug 15;98(4): 044314-11 pgs. http://dx.doi.org/10.1063/1.2008384 , Aug-2005